BROWN REPRESENTABILITY FOR EXACT FUNCTORS PRESERVING DIRECT SUMS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 11860-11876, following the statement at 11850-11858.

Editorial note. No omissions within this span: each step is argued or delegated to a named lemma whose statement is exactly the step.

Allowed standard prerequisites (editorial).

- Brown representability hypothesis. lemma-brown-bis (same chapter, cited at ~11871; its own span is gate-blocked by citations and used here only as a cited lemma). Produces the representing object X for the contravariant functor.
- Adjoint functor criterion. Categories, Lemma categories-lemma-adjoint-exists (cross-chapter, cited at ~11873). Objectwise representation yields the right adjoint.
- Exactness of adjoints. lemma-adjoint-is-exact (same chapter, cited at ~11875; cited for its statement). The right adjoint of an exact functor is exact.

Source reference. Original reference 0A8B: focused-seeds/geometry/S9-der-brown-bis/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0A8D: sources/stacks-derived/pinned-original-source.tex, source lines 2507–2507. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0GYG: sources/stacks-derived/pinned-original-source.tex, source lines 11686–11686. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Proposition 0.1. *Let $\mathcal{D}$ be a triangulated category with direct sums. Assume there exists a set $\mathcal{E}$ of objects of $\mathcal{D}$ satisfying conditions (1) and (2) of Lemma 0GYG. Let $F : \mathcal{D} \to \mathcal{D}'$ be an exact functor of triangulated categories which transforms direct sums into direct sums. Then F has an exact right adjoint.*

Date: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. For an object Y of $\mathcal{D}'$ consider the contravariant functor

$$\mathcal{D} \to Ab, \quad W \mapsto \mathrm{Hom}_{\mathcal{D}'}(F(W), Y)$$

This is a cohomological functor as F is exact and transforms direct sums into products as F transforms direct sums into direct sums. Thus by Lemma 0GYG we find an object X of $\mathcal{D}$ such that $\mathrm{Hom}_{\mathcal{D}}(W, X) = \mathrm{Hom}_{\mathcal{D}'}(F(W), Y)$. The existence of the adjoint follows from Categories, Lemma 0A8B. Exactness follows from Lemma 0A8D. $\square$